Similarity and normalisation

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

WORKBOOK · 1 OF 15

Recall and choose your starting point

> Record your starting explanation for Similarity and normalisation before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Embeddings and neighbourhoods

Without opening the old lesson, explain one key idea from Embeddings and neighbourhoods. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Augmentation and leakage

Without opening the old lesson, explain one key idea from Augmentation and leakage. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Convolution as a local operation

Without opening the old lesson, explain one key idea from Convolution as a local operation. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Lesson reference for workbook practice

Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

Last week, a neighbourhood depended on a chosen distance rule. This week, compare the same fictional vectors in two ways. Squared distance rewards nearby coordinates.

Cosine similarity compares direction and ignores positive changes in length. Neither rule is automatically right for every task.

Vector	Coordinates	Length
Query Q	[3, 4]	5
Candidate A	[6, 8]	10
Candidate B	[3, 0]	3

Vector	Coordinates	Length
Candidate C	[-3, -4]	5

For Q, length = square root of (3 squared + 4 squared) = square root of 25 = 5. Divide both components by 5 to get its **unit vector** [0.6, 0.8]. Its length is one because 0.6 squared + 0.8 squared = 1.

A becomes the same **unit vector**. C becomes [-0.6, -0.8], pointing in the opposite direction.

These supplied coordinates are for arithmetic practice. They are not learned language embeddings. Normalising each row keeps its direction but removes its original length unless that length is saved separately.

WORKBOOK · 2 OF 15

Practise with fading help

> Complete the supported Similarity and normalisation trace and record the first step you can explain without help.

Use W2: Compare the rankings for Similarity and normalisation. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 15

Individual reasoning check

> Check the proposed Similarity and normalisation explanation against the supplied facts before discussing it.

Use the worked case for Similarity and normalisation. Proposed explanation: “Cosine is a probability”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 15

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Compare a new case independently

Use query [5, 0] and candidates P [10, 0], R [4, 3] and S [0, 5]. Calculate all raw squared distances and cosines, name each winner, and give squared distances after normalising both query and candidates.

Explain the changed ranking.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 15

W1 Trace the unit vectors

> Trace the unit vectors. Apply the supplied Similarity and normalisation facts and keep your reasoning.

For Q [3, 4], A [6, 8] and C [-3, -4], give each length and **unit vector**. Explain one distinction that normalisation removes and one that it preserves.

My reasoning and evidence

WORKBOOK · 6 OF 15

W2 Compare the rankings

> Compare the rankings. Apply the supplied Similarity and normalisation facts and keep your reasoning.

For query Q, write the raw squared distances and cosines to A, B and C from the lesson. Name the winner under each rule. Calculate squared distances after normalising both sides and check the relationship to cosine.

My reasoning and evidence

WORKBOOK · 7 OF 15

W3 Handle a vector without direction

> Handle a vector without direction. Apply the supplied Similarity and normalisation facts and keep your reasoning.

A candidate set contains Z [0, 0] and B [3, 0]. With nonzero query Q, what happens under this lesson’s zero-vector policy? What if Q is zero, or Z is the only candidate?

Explain why replacing an undefined cosine with zero is misleading.

My reasoning and evidence

WORKBOOK · 8 OF 15

W4 Compare a new case independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Compare a new case independently. Apply the supplied Similarity and normalisation facts and keep your reasoning.

Use query [5, 0] and candidates P [10, 0], R [4, 3] and S [0, 5]. Calculate all raw squared distances and cosines, name each winner, and give squared distances after normalising both query and candidates.

Explain the changed ranking.

> **My reasoning and evidence**

W5 Audit a proposed improvement

> Audit a proposed improvement. Apply the supplied Similarity and normalisation facts and keep your reasoning.

For event-count vectors [3, 4] and [6, 8], what useful information can normalisation remove? A team chooses cosine because it improves results on the final evaluation set, then calls that same set independent **evidence**.

Explain how to repair the evaluation plan.

My reasoning and evidence

WORKBOOK · 10 OF 15

Project checkpoint and reflection

> Connect Similarity and normalisation to Image-pattern Network Investigation, identify a limit and choose the next useful test.

Add a labelled similarity and normalisation **evidence** sheet to Image-pattern Network Investigation. Show how you can calculate **vector length** and normalise a nonzero vector.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

WORKBOOK · 11 OF 15

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 12 OF 15

Pictorial case practice: Similarity and normalisation

> Compare angular alignment with raw distance on the same vectors. Explain the labelled step before checking the changed condition.

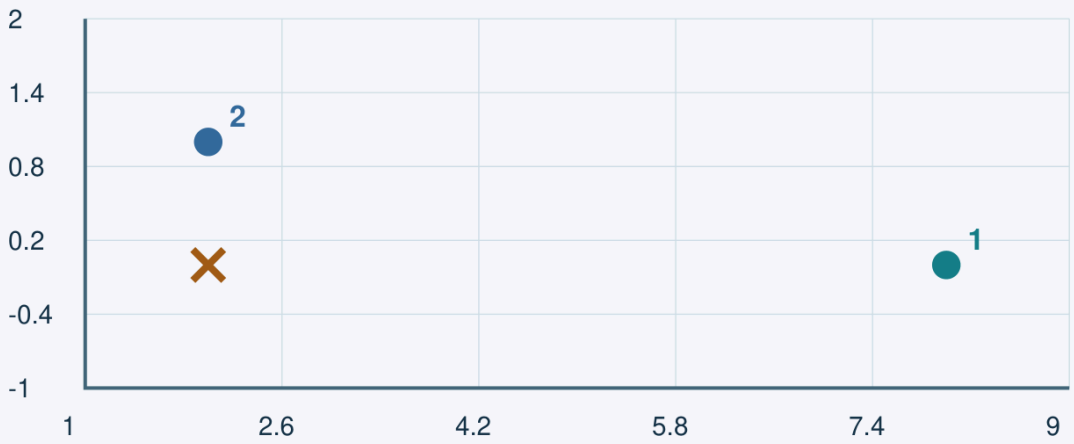
Compare angular alignment with raw distance on the same vectors.

Query [2, 0]; P=[8, 0]; Q=[2, 1]. Nonzero vectors only; compare squared Euclidean distance and cosine.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-5/27

Raw proximity differs from angular alignment



Horizontal: first vector coordinate; vertical: second vector coordinate

1: P at (8, 0)

2: Q at (2, 1)

Cross: query at (2, 0)

Distance rule: Compare squared Euclidean distance and cosine

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-5/27

Similarity and normalisation

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Raw distances

P distance 36; Q distance 1, so Q is closer in raw coordinates.

↓

2. Angular similarity

Cosine P =1; cosine Q =2/sqrt5≈0.8944, so P aligns more closely.

↓

3. Normalise

Unit query [1, 0], P [1, 0], Q [2/sqrt5, 1/sqrt5]; normalised distance to P is 0.

CHANGE ONE THING

Replace P with [0, 0].

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why do the original two metrics select different winners? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

WORKBOOK · 13 OF 15

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Image-pattern Network Investigation record.

Project: Image-pattern Network Investigation

Current milestone: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

WORKBOOK · 14 OF 15

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Compare angular alignment with raw distance on the same vectors.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Unit query [1, 0], P [1, 0], Q [2/sqrt5, 1/sqrt5]; normalised distance to P is 0.

1. Angular similarity

2. Normalise

3. Raw distances

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Normalise Unit query [1, 0], P [1, 0], Q [2/sqrt5, 1/sqrt5]; normalised distance to P is 0. Raw Euclidean distance includes differences in both direction and magnitude.

Cosine compares direction using the **vector length**s, so a long vector can align perfectly while being far away in raw coordinates. Declare which question your similarity measure answers.

A zero vector has no direction and cannot be divided by its norm for cosine or unit normalisation.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Replace P with [0, 0]. Which conclusion is supported by the supplied facts?

1. Cosine and unit normalisation for P are undefined.

2. This one example guarantees correct results for every future **input**.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Cosine and unit normalisation

for P are undefined. The zero vector has norm 0, so dividing by its norm is invalid.

Repair a missing building block

Review Embeddings and neighbourhoods (ai-5/26). Goal: Look up a vector using a token ID and preserve sequence shape. Short practice: What would justify claiming nearby vectors mean related topics?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Augmentation and leakage (ai-5/24). Goal: Check whether a transformation preserves the task label. Short practice: How many independent source groups are in the final test?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Convolution as a local operation (ai-5/21). Goal: Compute a local weighted sum at each valid image position. Short practice: Why is flipping the kernel an implementation change here?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

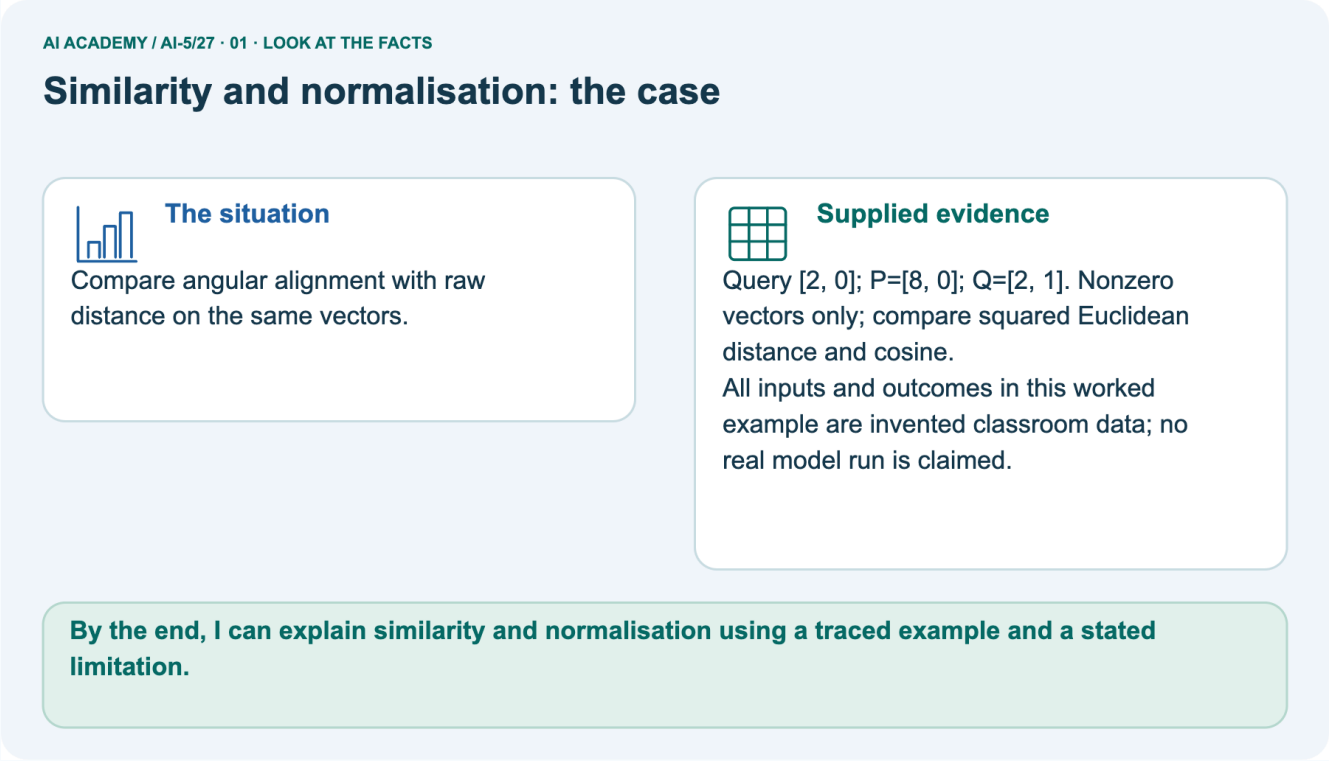
What I repaired, the evidence I kept and the next test

WORKBOOK · 15 OF 15

Visual concept journey

> Explain each Similarity and normalisation picture from its labels, then record a relevant limit.

01. Similarity and normalisation: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

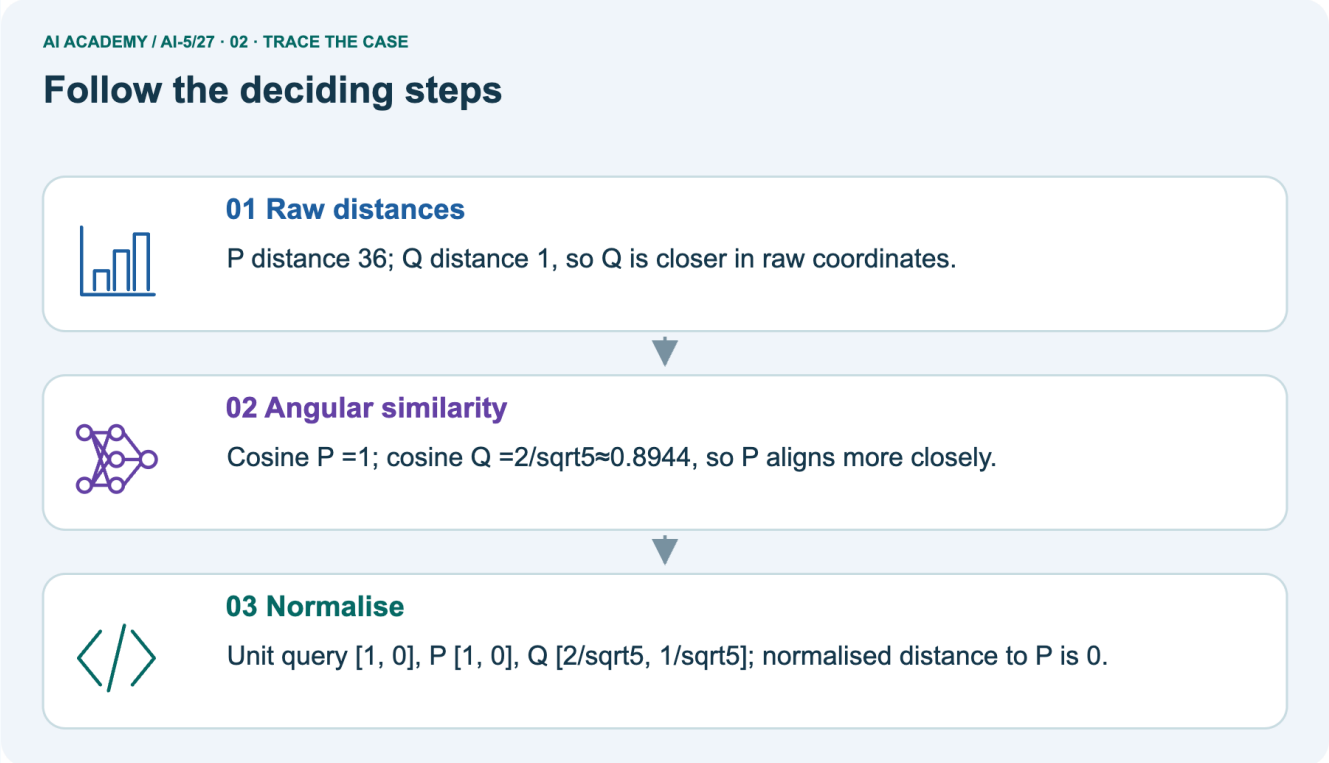
Compare angular alignment with raw distance on the same vectors. Supplied facts: Query [2, 0]; P=[8, 0]; Q=[2, 1]. Nonzero vectors only; compare squared Euclidean distance and cosine.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



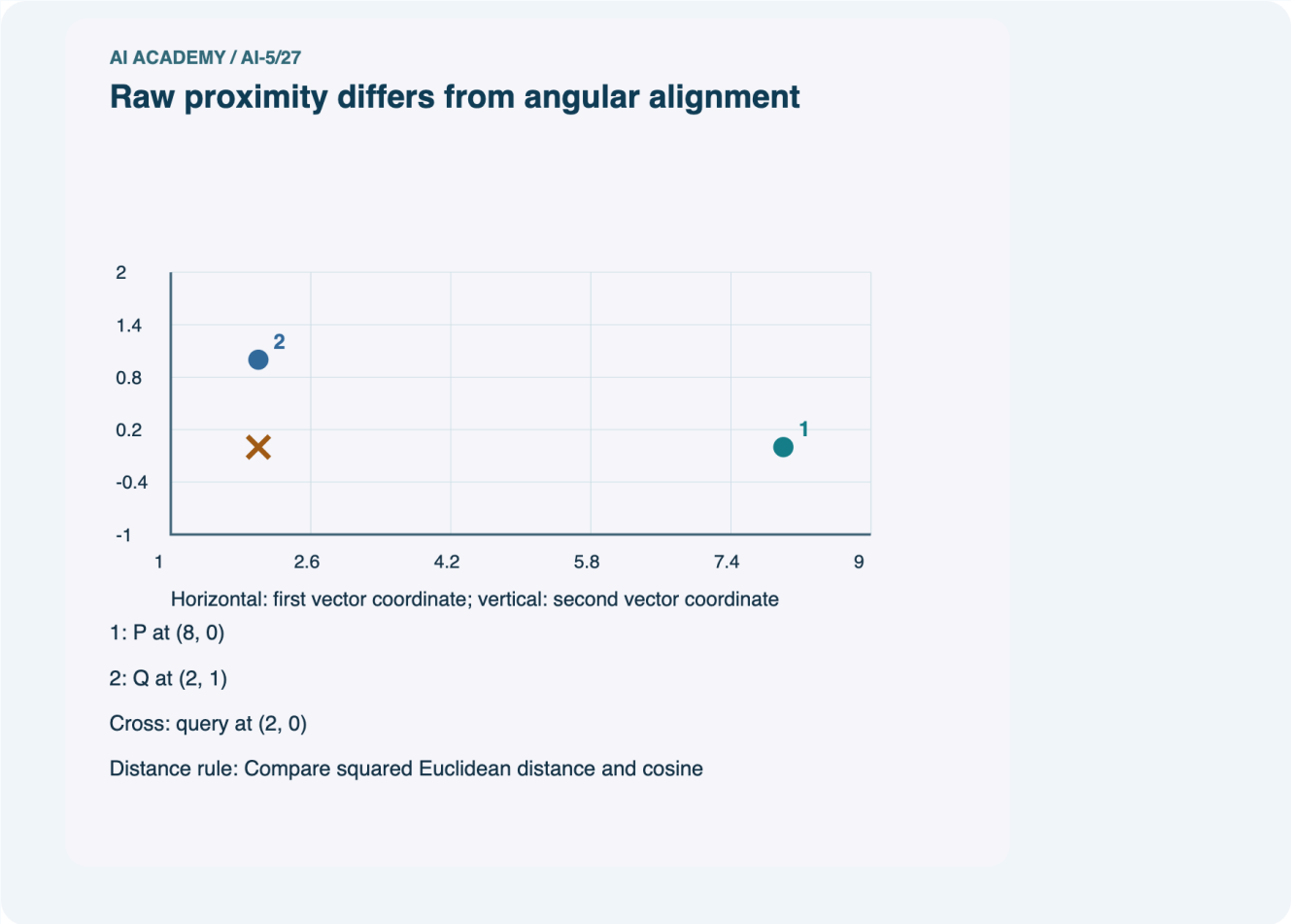
Follow the labelled picture with the matching explanation. Open the vector diagram

Raw distances: P distance 36; Q distance 1, so Q is closer in raw coordinates. → Angular similarity: Cosine P =1; cosine Q =2/sqrt5≈0.8944, so P aligns more closely. → Normalise: Unit query [1, 0], P [1, 0], Q [2/sqrt5, 1/sqrt5]; normalised distance to P is 0.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Raw proximity differs from angular alignment



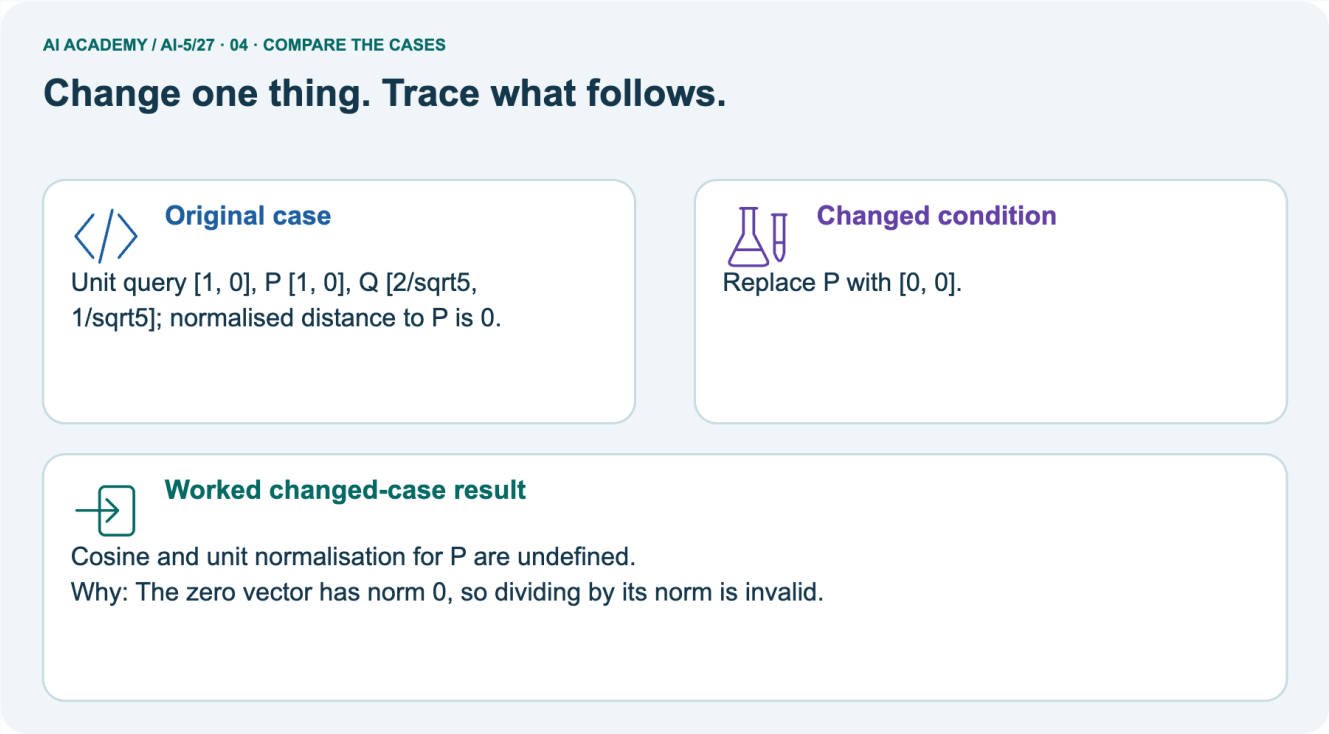
Follow the labelled picture with the matching explanation. Open the vector diagram

Raw proximity differs from angular alignment. P: (8, 0). Q: (2, 1). Query query: (2, 0). Axes: first vector coordinate; second vector coordinate. Distance: Compare squared Euclidean distance and cosine.

> Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

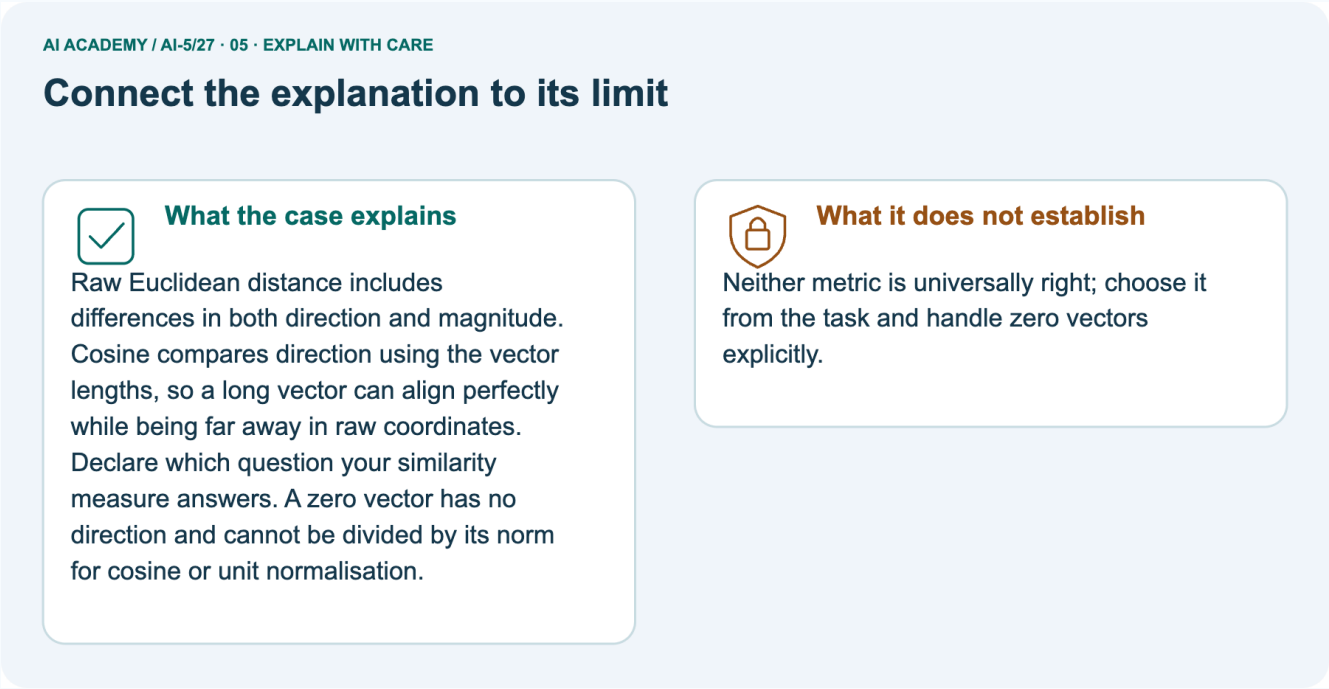
Original case: Unit query [1, 0], P [1, 0], Q [2/sqrt5, 1/sqrt5]; normalised distance to P is 0. Changed condition: Replace P with [0, 0]. Expected result: Cosine and unit normalisation for P are undefined.

Reason: The zero vector has norm 0, so dividing by its norm is invalid.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Raw Euclidean distance includes differences in both direction and magnitude. Cosine compares direction using the **vector length**s, so a long vector can align perfectly while being far away in raw coordinates.

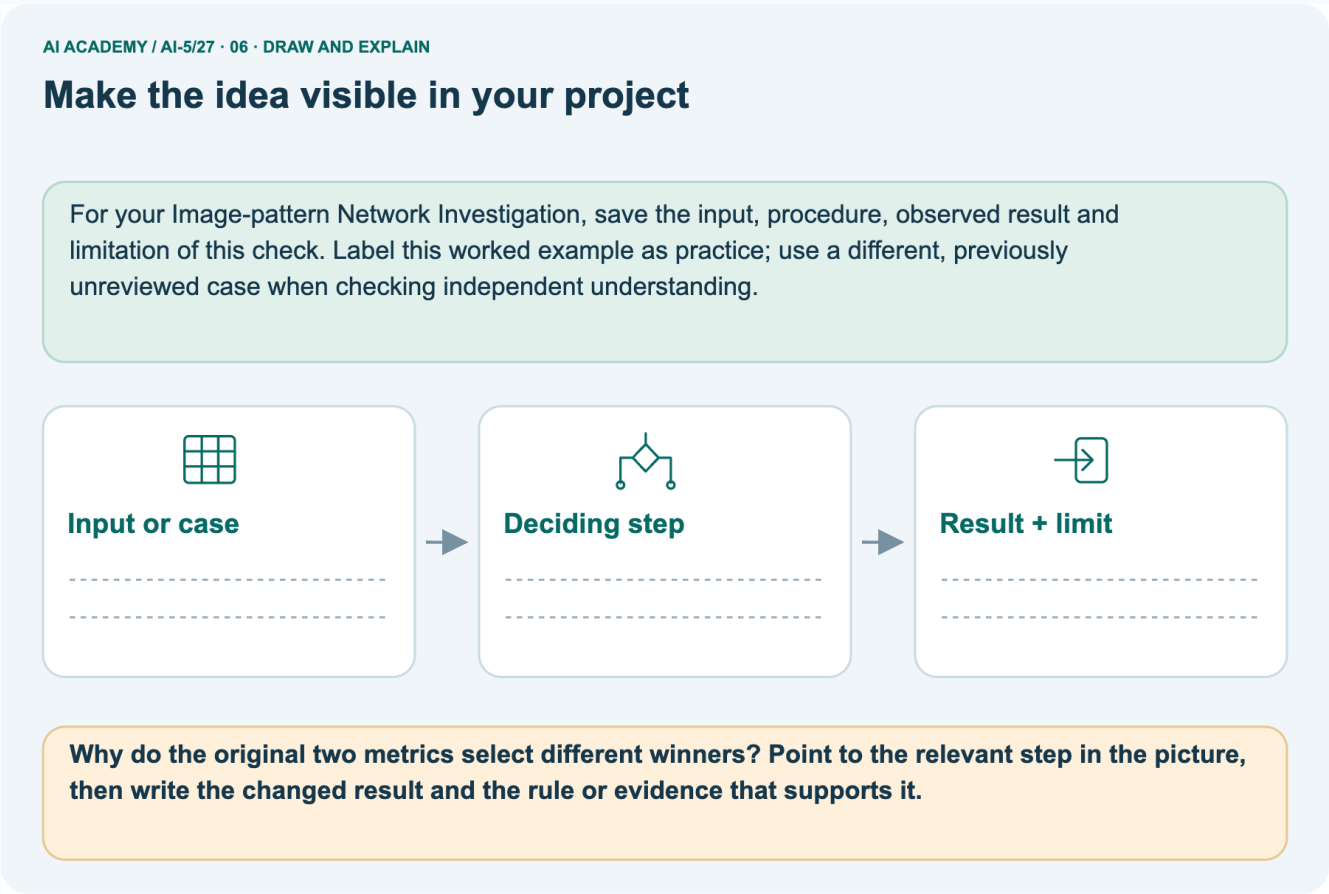
Declare which question your similarity measure answers. A zero vector has no direction and cannot be divided by its norm for cosine or unit normalisation.

Limit: Neither metric is universally right; choose it from the task and handle zero vectors explicitly.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why do the original two metrics select different winners? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.